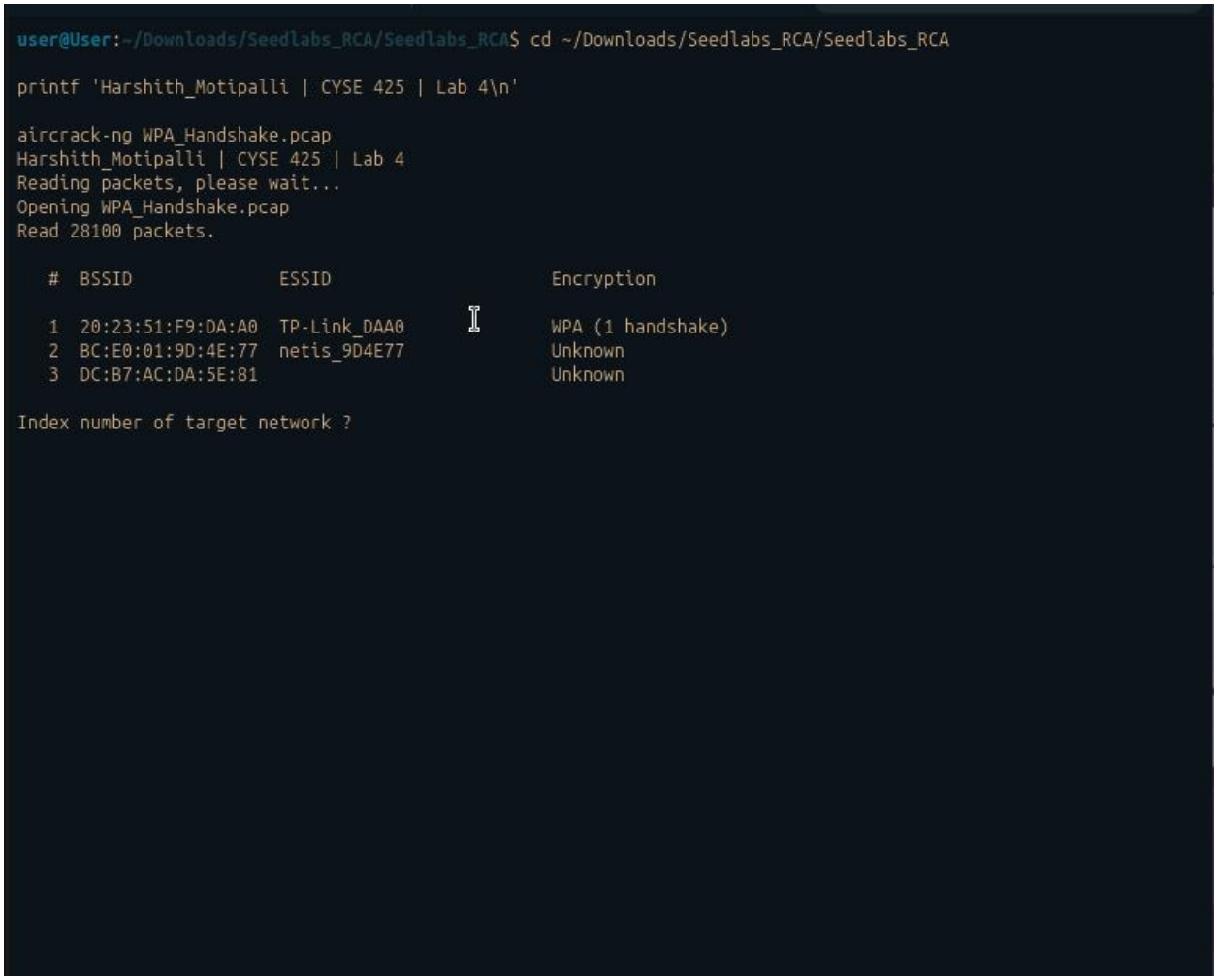
CYSE 425 - Lab 4

Harshith Motipalli
Professor Gebril

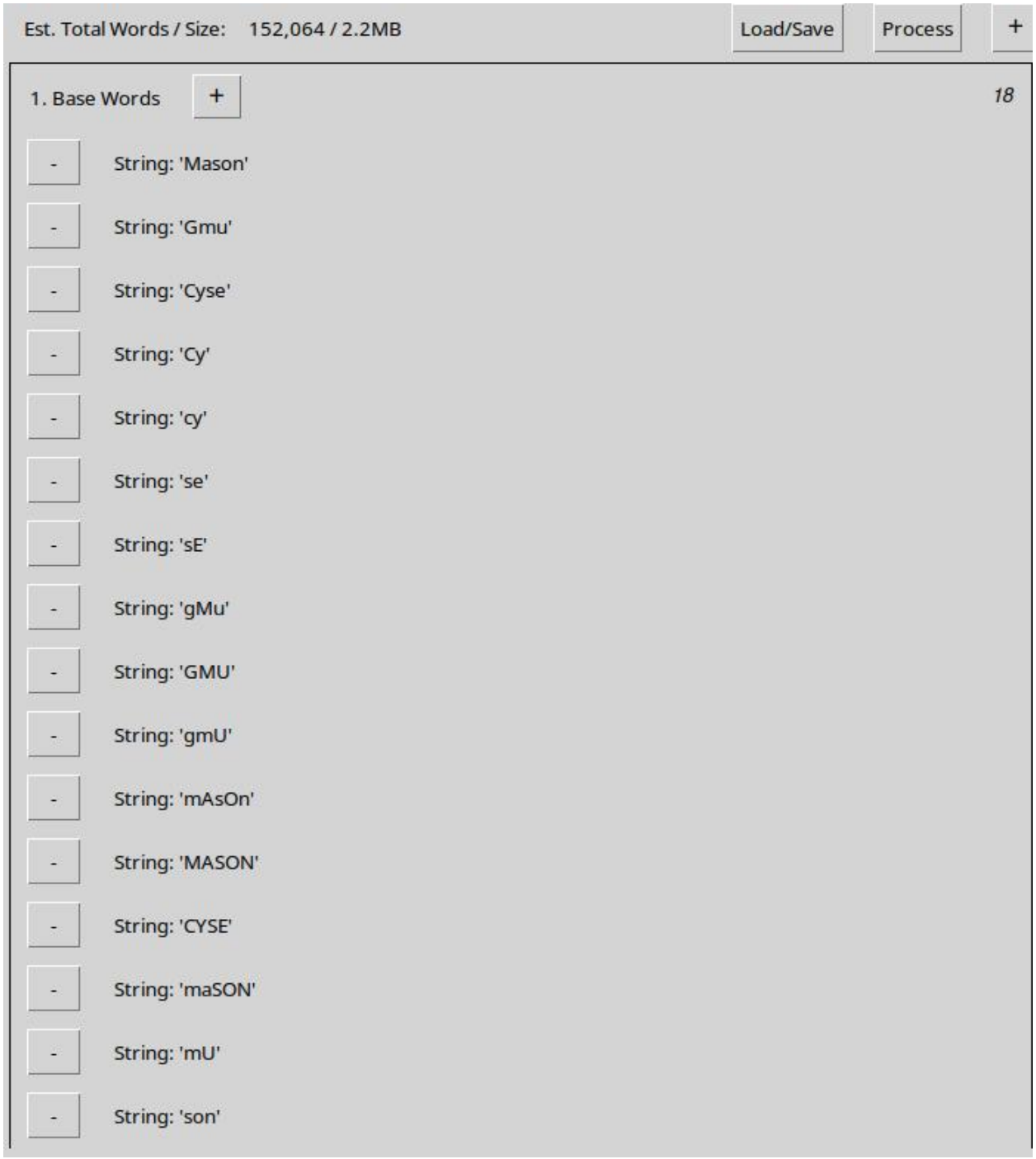
Purpose of Lab: The purpose of this lab was to understand how a WPA-protected wireless network can be vulnerable to password-cracking attacks. The lab involved analyzing a captured WPA four-way handshake, using Mentalist to generate a custom password wordlist based on specific requirements, and testing possible passwords to determine whether the network's pre-shared key could be recovered.

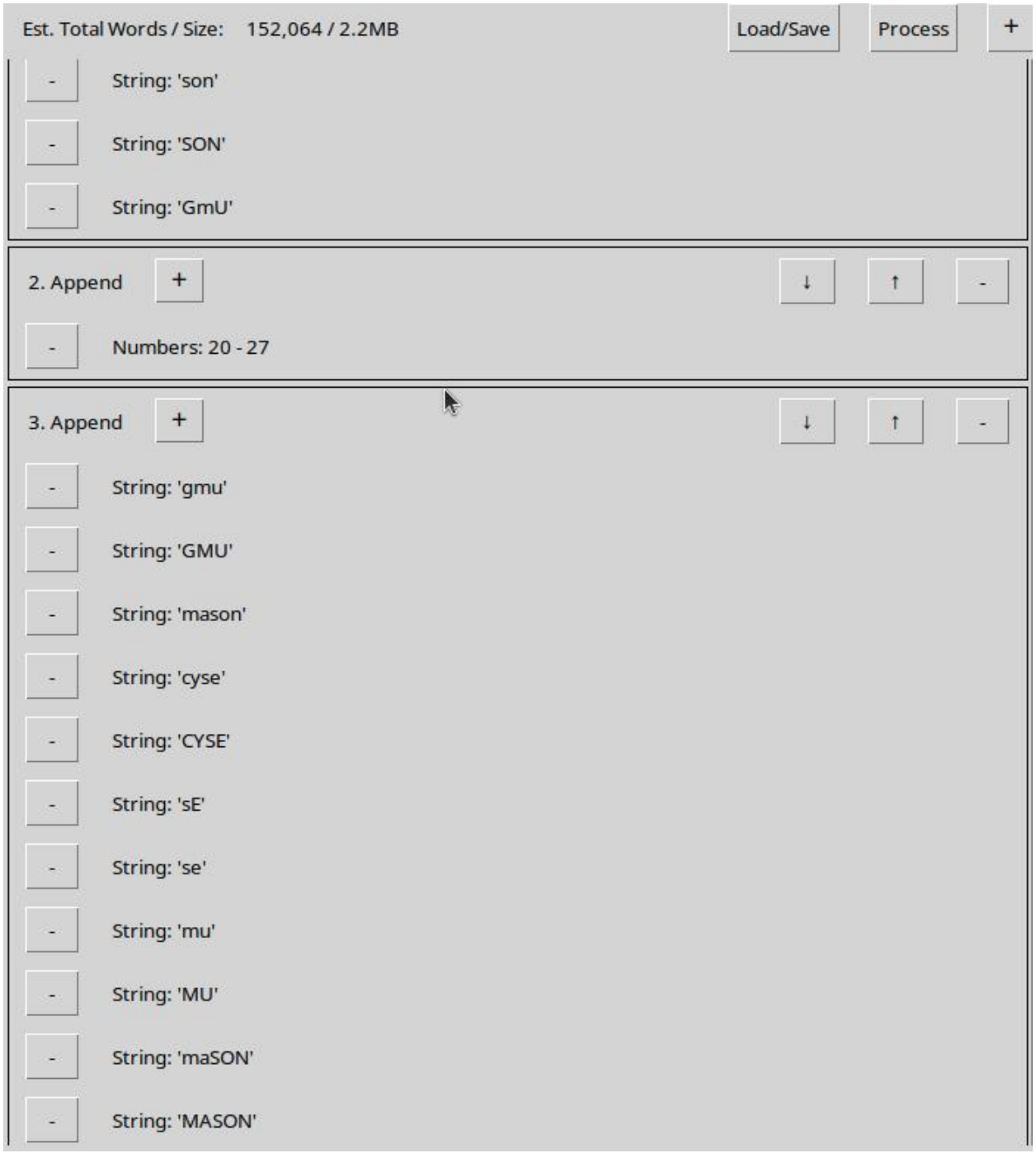
Task 1:

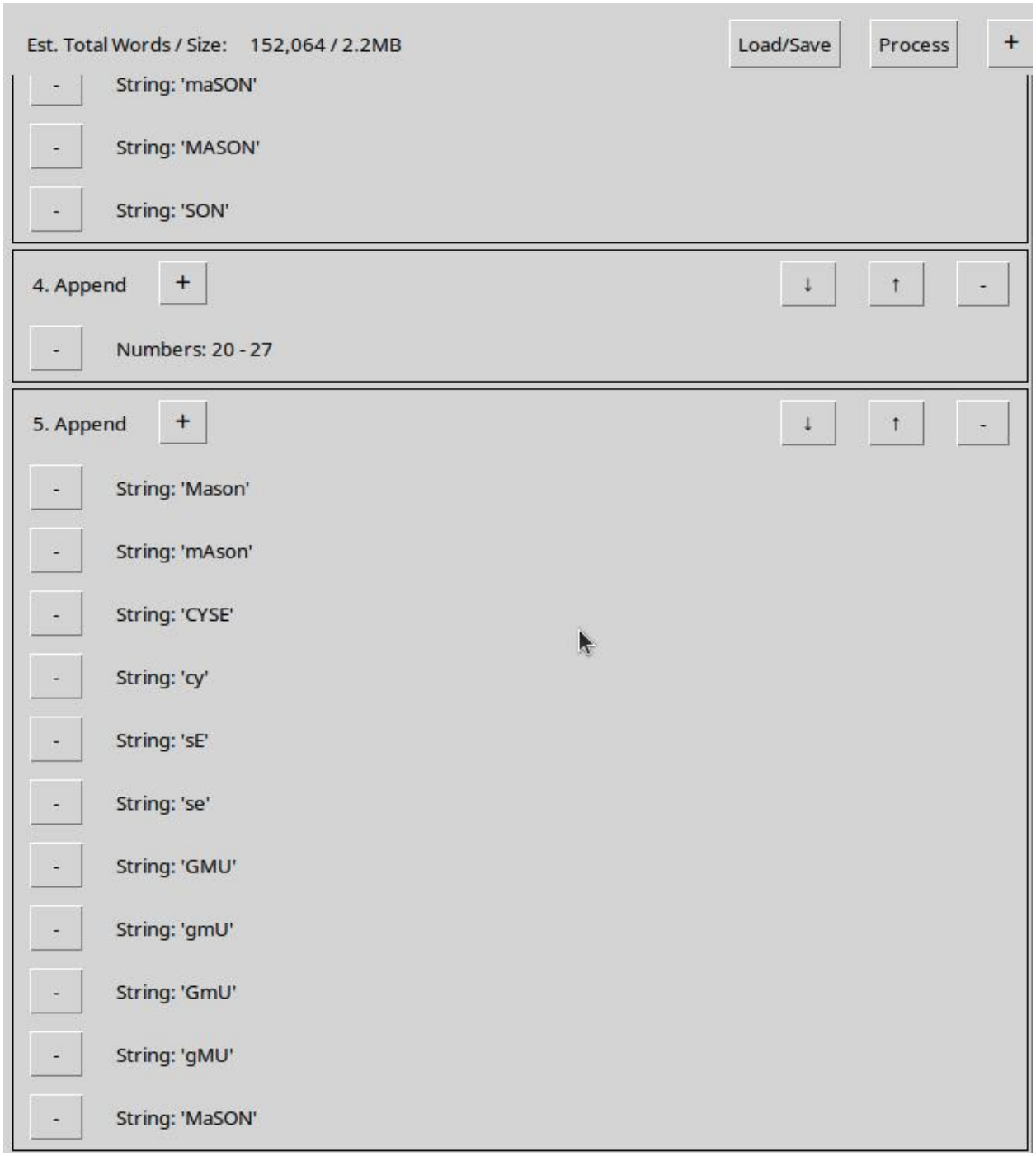


Description: I used Aircrack-ng to analyze the WPA_Handshake.pcap file and verify the presence of WPA four-way handshake. The tool processed 28,100 packets and identified the TP-Link_DAA0 network with BSSID 20:23:51:F9:DA:A0, confirming that one WPA handshake was captured. This verified that the file contained the authentication data necessary for offline password-cracking attempts.

Task 2:

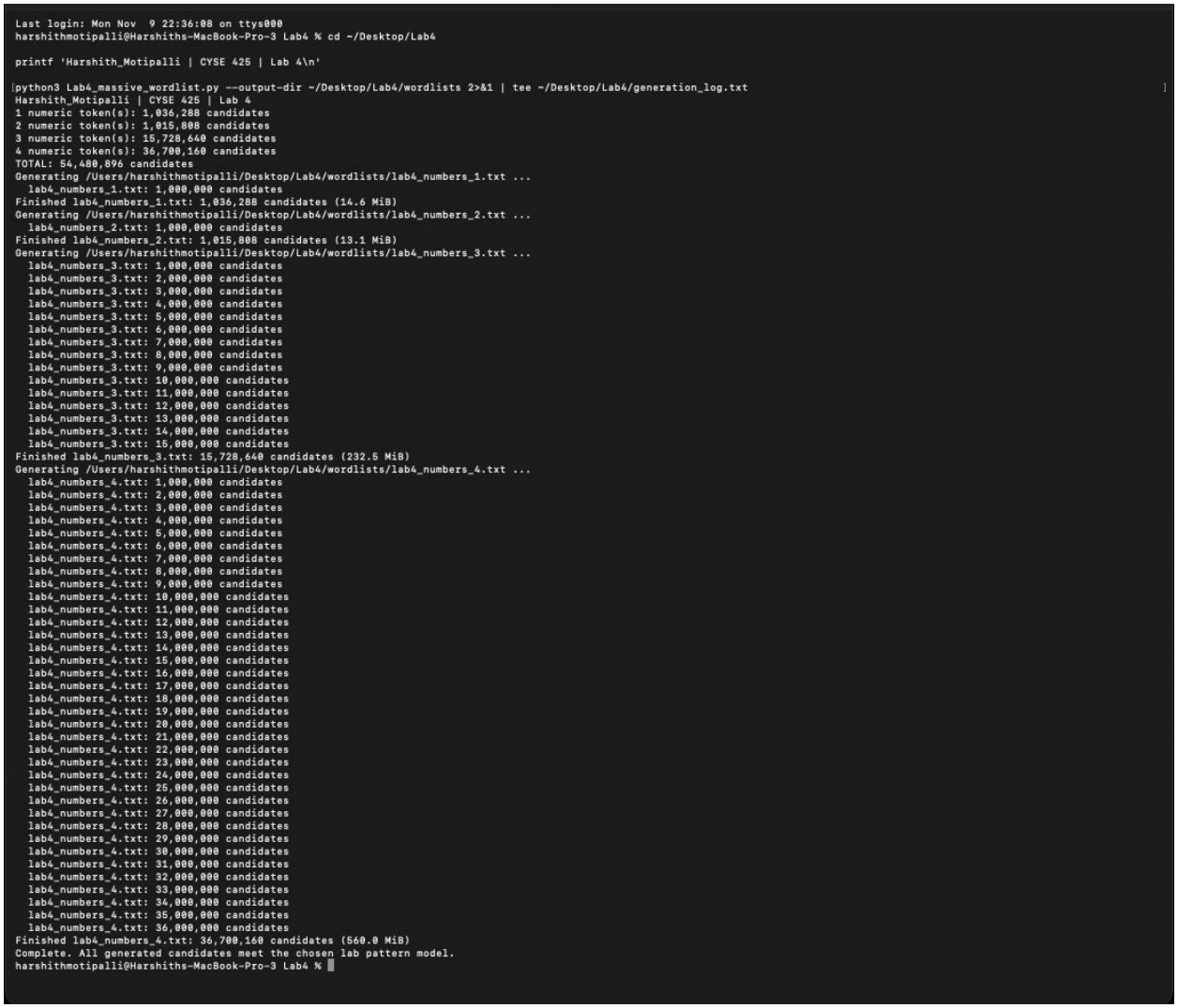

Est. Total Words / Size: 152,064 / 2.2MB
Load/Save
Process
+
1. Base Words
+
18
- String: 'Mason'
- String: 'Gmu'
- String: 'Cyse'
- String: 'Cy'
- String: 'cy'
- String: 'se'
- String: 'sE'
- String: 'gMu'
- String: 'GMU'
- String: 'gmU'
- String: 'mAsOn'
- String: 'MASON'
- String: 'CYSE'
- String: 'maSON'
- String: 'mU'
- String: 'son'


Est. Total Words / Size: 152,064 / 2.2MB
Load/Save
Process
+
-
String: 'son'
-
String: 'SON'
-
String: 'GmU'
2. Append
+
↓
↑
-
-
Numbers: 20 - 27
3. Append
+
↓
↑
-
-
String: 'gmu'
-
String: 'GMU'
-
String: 'mason'
-
String: 'cyse'
-
String: 'CYSE'
-
String: 'sE'
-
String: 'se'
-
String: 'mu'
-
String: 'MU'
-
String: 'maSON'
-
String: 'MASON'



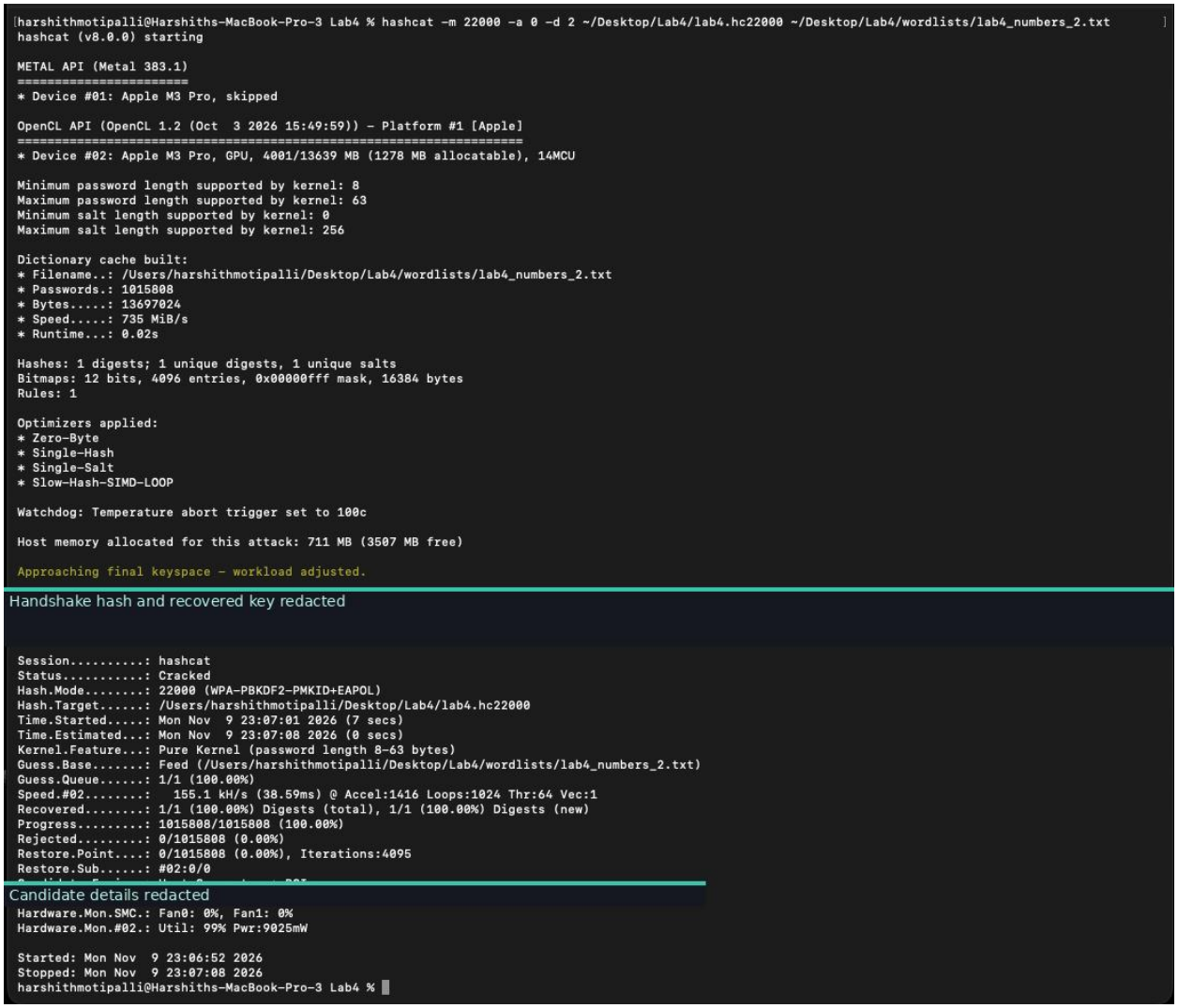
Description: I configured the Mentalist tool to generate password combinations using variations of MASON, CYSE, and GMU. Additionally, I added different uppercase and lowercase word fragments, followed by Append sections containing numbers from 20–27 and additional keyword combinations. This configuration generated an estimated 152,064 password candidates.

Task 3:



Description: I then used a Python script to generate a larger password wordlist based on the lab's requirements and the one built earlier on the Mentalist Tool. The script created combinations of the required keywords, different capitalization patterns, and multiple number placements. Four separate wordlists were generated, containing a total of 54,480,896 password candidates for testing with Hashcat.

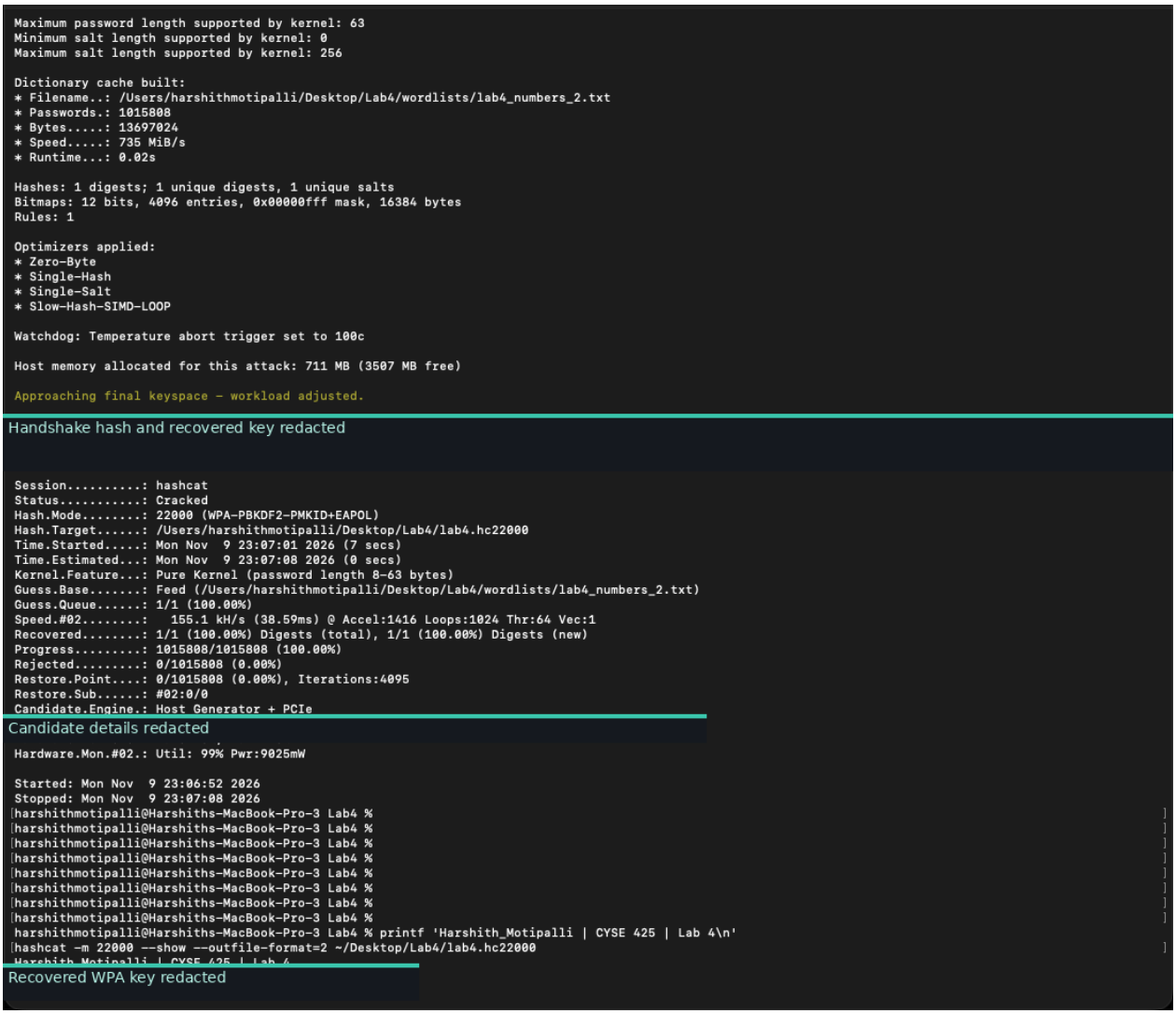
Task 4:



Description: Considering that my Ubuntu was processing the Hashcat keys at about 500 H/s, I used Hashcat 8.0.0 on my MacBook, which got 155.1 kH/s to test the generated password candidates against the captured WPA handshake. Hashcat used attack mode 0 and hash mode 22000 with the second generated wordlist, containing 1,015,808 candidates. The program successfully recovered the WPA password, displaying a status of "Cracked" and a recovery rate of 100%.

Task 5: Verify Password Recovery

Hashcat's verification output confirmed a recovered WPA pre-shared key.
The original key and handshake hash are redacted for public distribution.



Description: The offline WPA credential audit completed successfully,
and Hashcat verified a candidate matching the captured authentication data.
Sensitive credential material is not included in this public portfolio copy.